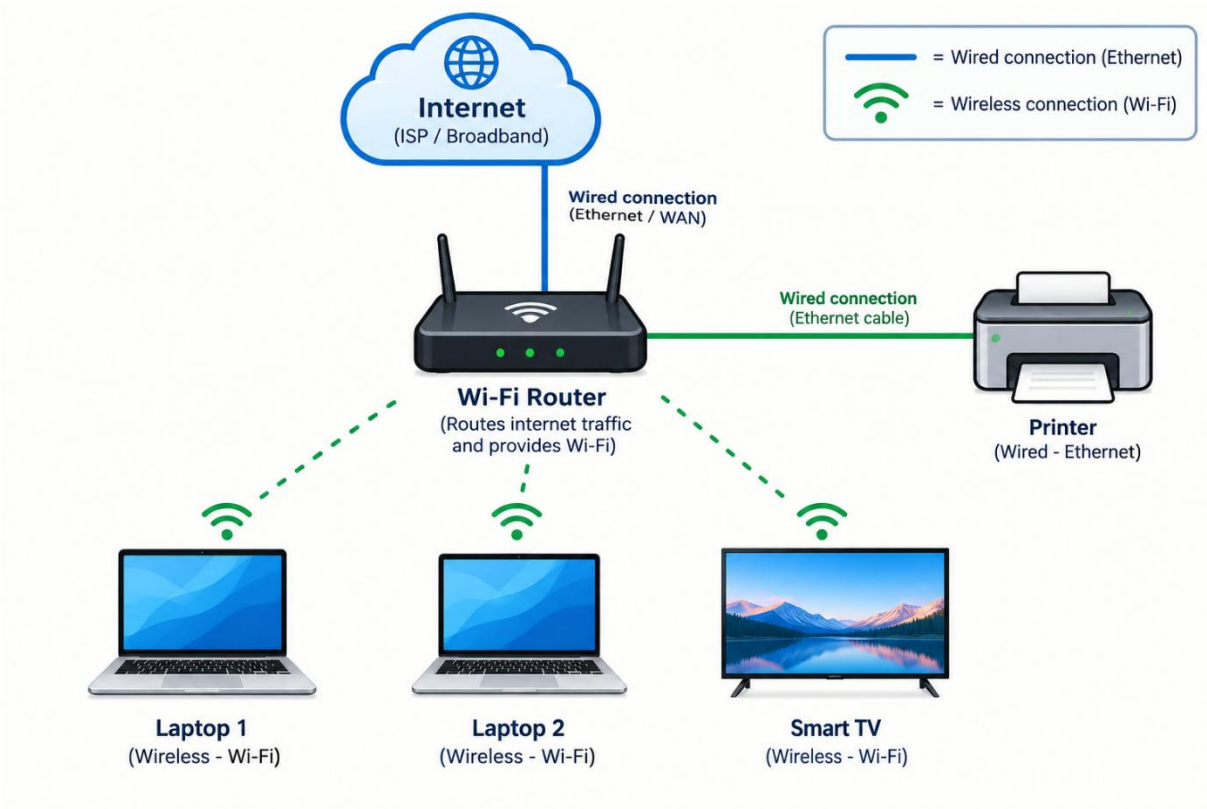
Assignment – 2.1

Build a SOHO Network

Q - 1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



Q - 2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

- **Hardware Components Required for SOHO Network:**

No.	Hardware component	Brand/model suggestion	Quantity
1	Broadband modem/ONT	JioFiber or Airtel Xstream Fiber	1
2	Wi-Fi router	TP-Link Archer C6	1
3	Laptop	Dell or Lenovo	2
4	Smart TV	Samsung or LG	1
5	Printer	HP DeskJet or HP Smart Tank	1
6	Ethernet cable (Cat6)	D-Link	2–3

No.	Hardware component	Brand/model suggestion	Quantity
7	Network switch	TP-Link TL-SG105	1 (optional)
8	Wi-Fi extender	TP-Link RE315	1 (optional)
9	Power cables/adapters	Compatible with each device	As required

➢ Connection details:

1. Connect the broadband modem to the Wi-Fi router using an Ethernet cable.
2. Connect both laptops to the router using Wi-Fi.
3. Connect the smart TV to the router using Wi-Fi.
4. Connect the printer using Wi-Fi or USB, depending on the printer model.
5. Use a network switch if extra wired ports are needed.
6. Use a Wi-Fi extender if the signal is weak.

Q - 3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

➢ Steps to Configure a Wi-Fi Router:

1. **Connect the router:** Connect the broadband cable to the router and turn it on.
2. **Connect a laptop:** Connect your laptop to the router using Wi-Fi or a LAN cable.
3. **Open router settings:** Open a browser and enter 192.168.0.1 or 192.168.1.1.
4. **Log in:** Enter the default username and password.
5. **Change Wi-Fi name:** Set a unique SSID, such as MyOffice_WiFi.
6. **Enable security:** Select **WPA2-PSK** in the Wi-Fi security settings.
7. **Set Wi-Fi password:** Create a strong password.
8. **Change admin password:** Replace the default router admin password with a new one.
9. **Save settings:** Click Save or Apply and restart the router if needed.

10. **Connect devices:** Connect both laptops, the smart TV, and the printer to the network.
11. **Test the internet:** Check that all devices are working properly.

- **Result:**

  The Wi-Fi router is configured with a unique name, a secure password, and a new admin password.

Q - 4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

- **IP Address Configuration for SOHO Network:**

  Use the IP addresses shown below for the SOHO network.

- **Router IP:** 192.168.1.1

Device	MAC Address (Example)	IP Address	Type
Wi-Fi Router	00:11:22:33:44:01	192.168.1.1	Static
Laptop 1	00:11:22:33:44:02	192.168.1.100	DHCP
Laptop 2	00:11:22:33:44:03	192.168.1.101	DHCP
Smart TV	00:11:22:33:44:04	192.168.1.10	Static
Printer	00:11:22:33:44:05	192.168.1.20	Static

Q - 5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

- **Troubleshooting Laptop Internet Connection:**

If a laptop cannot connect to the internet, I would check these problems:

No.	Problem	Solution
1	Wi-Fi is turned off	Turn on Wi-Fi and connect to the correct network.
2	Wrong Wi-Fi password	Enter the correct password.
3	Router problem	Restart the Wi-Fi router.
4	Incorrect IP address	Set the laptop to obtain an IP address automatically (DHCP).
5	Weak Wi-Fi signal	Move the laptop closer to the router.